



LOAN APPROVAL PREDICTION SYSTEM

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INTRODUCTION

OBJECTIVE:

- **EXPLORATORY DATA ANALYSIS** helps to look at data before making any assumptions.
- **K-NN CLASSIFICATION** classifies objects with learning data that are closest to the objects.
- **K-MEANS CLUSTERING** minimizes the total distance between points and their assigned cluster centroid.

DATASET OVERVIEW:

- The given dataset is about the prediction of bank loan approval (302 identifiers)
- No.of records - 13 (Applicant income, loan amount, credit history, loan status and etc.)

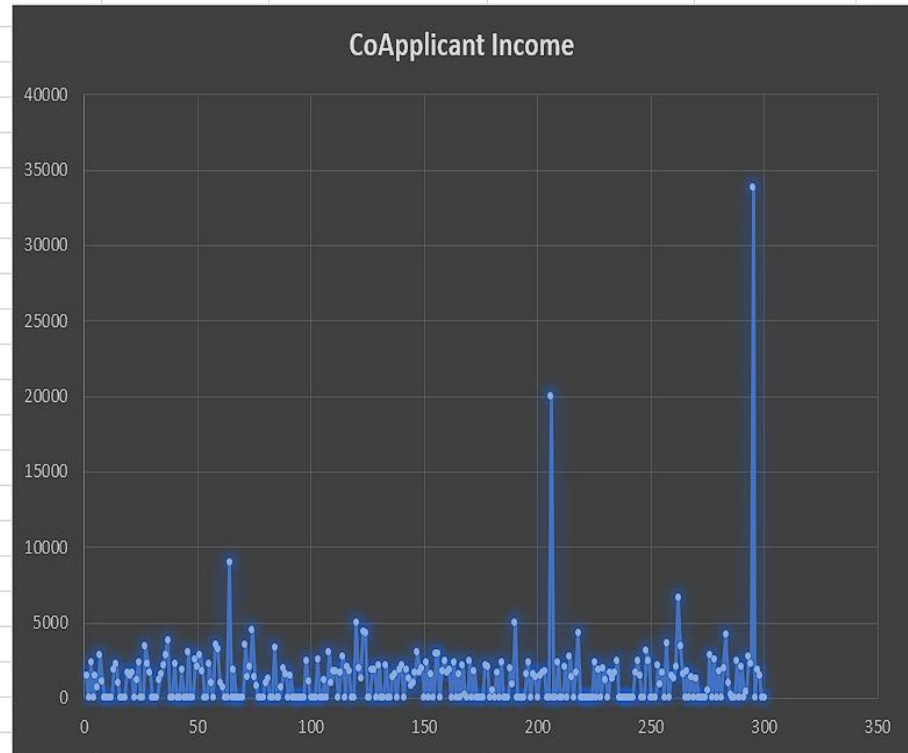
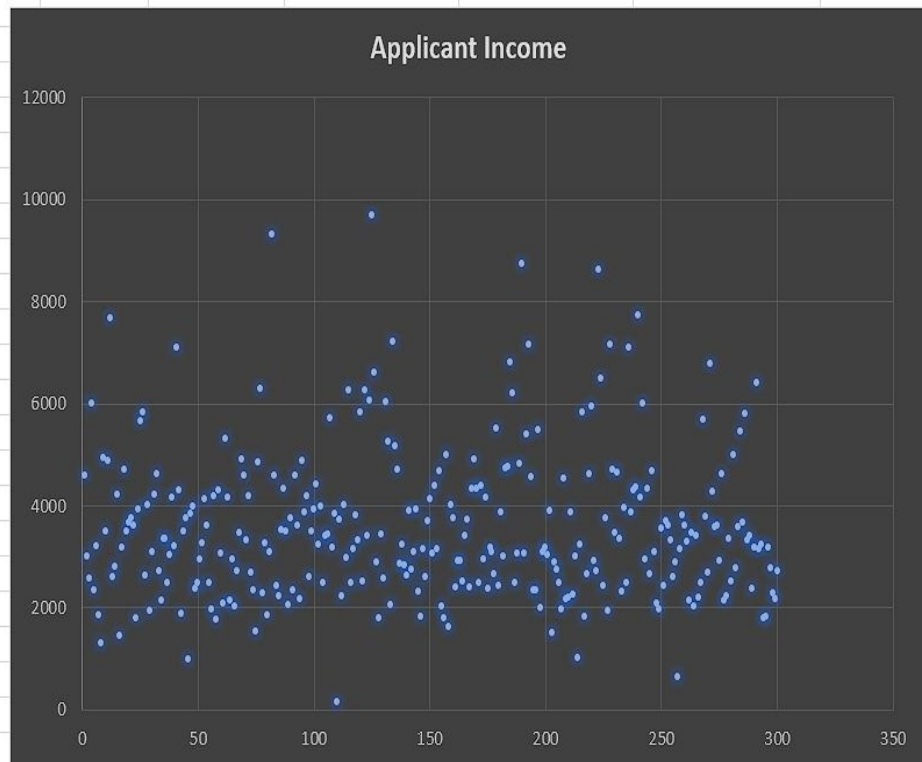
TOOLS AND TECHNIQUE:

- EXCEL
- Google slides

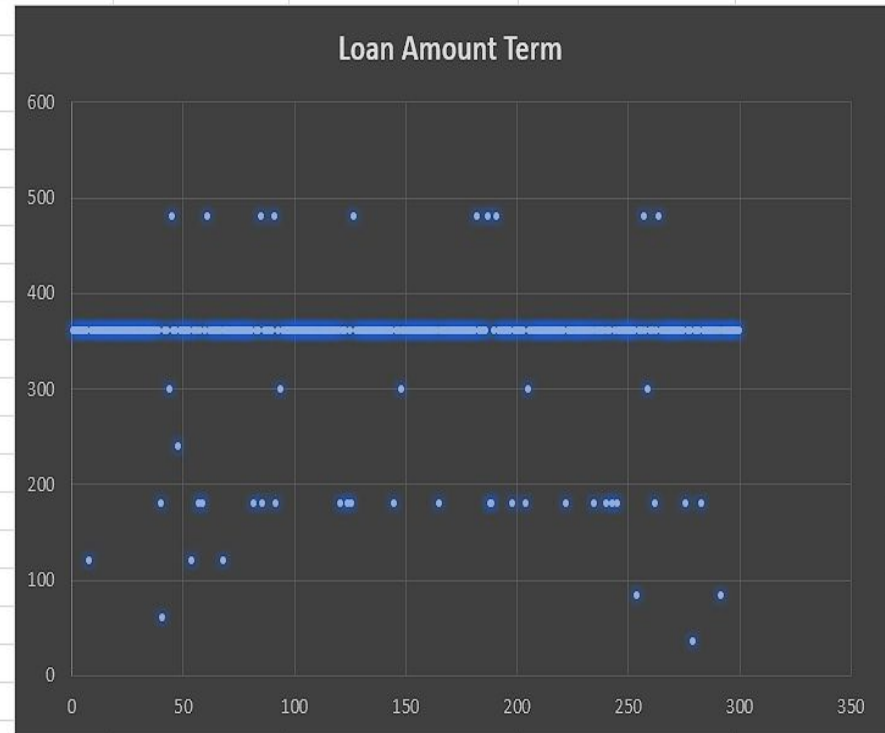
EXPLORATORY DATA ANALYSIS (EDA)

[illegible]

VISUAL REPRESENTATION:



VISUAL REPRESENTATION:



METHODOLOGY

K-NN CLASSIFICATION

Distance Calculation: *It is calculated by taking the square root of the sum of squared differences between the corresponding coordinates of the two points.*

Selecting Nearest Neighbours:

Step 1: *Select the number K of the neighbours.*

Step 2: *Calculate the euclidean distance of K number of neighbours.*

Step 3: *Take the K nearest neighbours as per the calculated euclidean distance.*

Step 4: *Among these K neighbours, count the number of the data points in each category.*

Predicting outcome:

- *The distance between the new point and each training point is calculated.*
- *The closest K data points are selected.*
- *The average of these data points is the final prediction of the new point.*

K-MEANS CLUSTERING:

K-Means Clustering:

- *The random partition method first randomly assigns a cluster to each observation and then proceeds to the updated step.*
- *Assigns data points to one of the K clusters depending on their distance from the centre of the clusters.*
- *Each vector gets added to a centroid once, so the complexity of the step is as followed in the result.*

K-NN CLASSIFICATION RESULTS:

[illegible]

K-MEANS CLUSTERING RESULT:

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Loan_ID ▾	Gender ▾	Married ▾	Dependent ▾	Education ▾	Self_Employed ▾	Applicant_Income ▾	Coapplicant_Income ▾	Loan_Amount ▾	Loan_Amount_Term ▾	Credit_History ▾	Property_Area ▾	Loan_Status ▾
283	LP002777	Male	Yes	0	Graduate	No	2785	2016	110	360	1	Rural	Y
284	LP002789	Male	Yes	0	Graduate	No	3593	4266	132	180	0	Rural	N
285	LP002792	Male	Yes	1	Graduate	No	5468	1032	26	360	1	Semiurban	Y
286	LP002807	Male	Yes	2	Not Graduate	No	3675	242	108	360	1	Semiurban	Y
287	LP002821	Male	No	0	Not Graduate	Yes	5800	0	132	360	1	Semiurban	Y
288	LP002836	Male	No	0	Graduate	No	3333	0	70	360	1	Urban	Y
289	LP002837	Male	Yes	3+	Graduate	No	3400	2500	123	360	0	Rural	N
290	LP002840	Female	No	0	Graduate	No	2378	0	9	360	1	Urban	N
291	LP002841	Male	Yes	0	Graduate	No	3166	2064	104	360	0	Urban	N
292	LP002863	Male	Yes	3+	Graduate	No	6406	0	150	360	1	Semiurban	N
293	LP002868	Male	Yes	2	Graduate	No	3159	461	108	84	1	Urban	Y
294	LP002874	Male	No	0	Graduate	No	3229	2739	110	360	1	Urban	Y
295	LP002877	Male	Yes	1	Graduate	No	1782	2232	107	360	1	Rural	Y
296	LP002893	Male	No	0	Graduate	No	1836	33837	90	360	1	Urban	N
297	LP002894	Female	Yes	0	Graduate	No	3166	0	36	360	1	Semiurban	Y
298	LP002911	Male	Yes	1	Graduate	No	2787	1917	146	360	0	Rural	N
299	LP002916	Male	Yes	0	Graduate	No	2297	1522	104	360	1	Urban	Y
300	LP002917	Female	No	0	Not Graduate	No	2165	0	70	360	1	Semiurban	Y
301	LP002926	Male	Yes	2	Graduate	Yes	2726	0	106	360	0	Semiurban	N
302	MAXIMUM	0	0	2	0	0	9703	33837	150	480	1	NIL	NIL
303	MINIMUM	0	0	0	0	0	150	0	9	36	0	NIL	NIL
304	MEDIAN	NIL	NIL	0	NIL	NIL	3321	918	110	360	1	NIL	NIL
305	AVERAGE	NIL	NIL	0.492805755	NIL	NIL	3591.05	1283.639733	104.7866667	342.48	0.8533333333	NIL	NIL

LEARNING AND INSIGHTS

Highlight how the analysis tools or machine learning techniques contributed to solving the problem:

There are two insightful methods to guide your machine learning models depending on what data is available and what question is implied, the algorithm will be trained to generate an outcome using one of these models.

- ❖ Supervised learning and
- ❖ Unsupervised learning.

Summary:

The graph of their patterns are particularly like wave graphs but the visibility is different from other graphs because, at only one point the wave of the graph is increasing.

CHALLENGE AND RECOMMENDATION

Challenges:

There is no appropriate challenges faced and I assuredly loved the organised journey of Artificial Intelligence and Machine Learning.

Recommendations:

To provide a clear instructed idea about dataset in home projects to the students via demo models.

CONCLUSION AND REFERENCE

Conclusion:

The course was well updated and provided concise information about Artificial Intelligence and Machine Learning.

The objective of the workshop has implanted the interest to pursue Data Science and Machine Learning has my career field.

Reference:

- *IITM Workshop sessions*
- *Youtube*
- *Google*